STATISTICS AND PROBABILITY

CHAPTER 12

Probability

12

SYLLABUS 4.5 4.6 4.11 4.13

Probability puts a number on how likely an event is. That single step is what turns risk into something you can measure, compare and act on, so that a decision can rest on more than a feeling about what might happen.

Here is a decision that turns on such a number. A health service screens everyone in an age group for a rare disease. The test is a good one. Of every 100 people who have the disease, it correctly identifies 99. Of every 100 people who do not have it, it correctly clears 95. About one person in a hundred has the disease.

Your result comes back positive. How likely is it that you have the disease?

Most people answer about 99%, and doctors asked the same question often answer the same way. The correct answer is about 17%. The test is not faulty and none of the numbers are unusual. The error is in the reasoning. That 99% says how the test behaves *when the disease is present*, and the quantity you actually want is a different one: how likely the disease is *now that the result is positive*. Those two numbers are not the same, and this chapter builds the rule that connects them.

So probability gives clear answers, and it is easy to apply a clear answer to the wrong question. Two habits guard against that, and both are practised throughout the chapter. The first is to state precisely which event you are asking about. The second is to keep track of where a probability came from: some are calculated from the symmetry of a fair die, others only measured by counting what happened in 200 trials, and the two are not equally trustworthy.

We begin with the simplest experiment there is, a single roll of a die. When we return to the test in the last section, the answer will take three lines.

12.1 The Language of Probability

Probability measures how likely something is, on a scale from 0 (impossible) to 1 (certain). To compute it we need to be precise about what "something" means.

Five words are used precisely from here on. The die is used to illustrate each one.

- **Experiment** — any process with an uncertain result. Rolling a die is an experiment.
- **Trial** — one repetition of the experiment. One roll is one trial.
- **Outcome** — the result of a single trial, such as a 3.
- **Sample space** U — the set of all possible outcomes. For one die, $U = \{1, 2, 3, 4, 5, 6\}$.
- **Event** — any subset of the sample space, that is, a collection of outcomes you are interested in. "An even number" is the event $\{2, 4, 6\}$.

When every outcome is equally likely, probability is just counting: the fraction of outcomes that belong to the event.

KEY · IB

$$P(A) = \frac{n(A)}{n(U)}$$

Here $n(A)$ is the number of outcomes in event A, and $n(U)$ the number in the whole sample space. Every probability obeys $0 \leq P(A) \leq 1$.

EXAMPLE 12.1

A fair die is rolled once. Find the probability of (a) an even number, (b) a number greater than 4.

The sample space has $n(U) = 6$ equally likely outcomes.

$$P(\text{even}) = \frac{n(\{2,4,6\})}{6} = \frac{3}{6} = \frac{1}{2}, \qquad P(>4) = \frac{n(\{5,6\})}{6} = \frac{2}{6} = \frac{1}{3}.$$

Counting works only because the die is fair. If the outcomes were not equally likely, this formula would not apply.

Relative frequency

When the outcomes are not equally likely, counting fails, and the probability has to be estimated by experiment. Repeat the trial many times and record how often the event occurs. The fraction of trials in which it happens is the **relative frequency** of the event, and it serves as an experimental estimate of the probability.

KEY After n trials,

$$P(A) \approx \frac{\text{number of trials in which } A \text{ occurred}}{n}$$

The estimate is rough when n is small, and it becomes more stable as n grows. This is the difference between the two kinds of probability you will meet: a *theoretical* probability comes from the structure of the experiment (a fair die, by symmetry), while an *experimental* one comes from data, and is what you fall back on when there is no symmetry to use.

EXAMPLE 12.2

A drawing pin is dropped 200 times and lands point-up 78 times. Estimate the probability that a dropped pin lands point-up.
There is no symmetry argument for a drawing pin, so use the relative frequency:

$$P(\text{point-up}) \approx \frac{78}{200} = 0.39.$$

A further 200 drops would almost certainly not give exactly 78 again, but the estimate becomes more accurate as the number of trials increases.

The complement

Sometimes the event you want is awkward to count, but its opposite is easy. The **complement** of A, written A', is the event that A does *not* happen. Since A either happens or it does not, their probabilities must fill the whole scale.

KEY · IB

$$P(A) + P(A') = 1 \qquad \Longrightarrow \qquad P(A') = 1 - P(A)$$

The complement is worth trying whenever the event itself is hard to count. In particular, whenever a question contains the phrase "at least one," work out the complement first.

EXAMPLE 12.3

A fair coin is tossed three times. Find the probability of getting at least one head.
Counting every way to get one, two, or three heads is possible but slow. The complement of "at least one head" is "no heads at all," a single outcome: TTT.

$$P(\text{at least one head}) = 1 - P(\text{TTT}) = 1 - \frac{1}{8} = \frac{7}{8}.$$

One subtraction replaces a count of seven outcomes. The saving becomes far greater as the number of tosses increases.

Expected number of occurrences

If an event has probability p and you repeat the experiment n times, you expect it to happen about np times.

KEY If an event has probability p and the experiment is repeated n times, the **expected number of occurrences** is

$$\text{expected number} = np.$$

This is not a guarantee. It is a long-run average. If you repeated the whole set of n trials many times and averaged the counts, that average would be close to np.

EXAMPLE 12.4

A fair die is rolled 60 times. How many sixes would you expect?

Each roll gives a six with probability $\frac{1}{6}$, so over 60 rolls:

$$\text{expected sixes} = 60 \times \frac{1}{6} = 10.$$

You will rarely get exactly 10, but the average over many sets of 60 rolls will be close to 10.

YOUR TURN 12.1 The Language of Probability

1.

(a) A fair die is rolled. Find the probability that the number is odd.
(b) A card is drawn from a deck of 52. Find the probability that it is a heart.
(c) A bag holds 4 red, 6 green and 5 blue counters. Find the probability that a counter drawn at random is green.
(d) A letter is chosen at random from the word STATISTICS. Find the probability that it is an S.
(e) A spinner numbered 1 to 8 is spun. Find the probability that it lands on a multiple of 3.

2.

(a) Given $P(A) = 0.35$, find $P(A')$.
(b) A fair die is rolled 90 times. Find the expected number of 5s.
(c) Two fair coins are tossed. Find the probability of at least one tail.

3.

(a) Two fair coins are tossed. List the sample space, and find the probability of exactly one head.
(b) A spinner has five sectors numbered 1 to 5, but it is not fair. In 400 spins it lands on 3 exactly 96 times. Estimate the probability of landing on 3.
(c) Using that estimate, find the expected number of 3s in 250 spins.
(d) Explain why the probability in part (a) can be found without any experiment, while the one in part (b) cannot.

4.

(a) A bag holds 12 counters, 5 of which are red. Find the probability that a counter drawn at random is not red.
(b) The probability that a bus is late on a given day is 0.18. Find the probability that it is not late.
(c) A fair coin is tossed four times. Find the probability of at least one tail.

12.2 Combining Events

Real questions rarely involve a single event. You want the probability that a student plays football *or* tennis, or that a card is a king *and* a heart. Two operations cover every case: the **union** $A \cup B$ ("A or B or both") and the **intersection** $A \cap B$ ("A and B together").

The word *or* needs care, because everyday English and mathematics do not use it the same way. "Tea or coffee?" offers you one drink, not both. In mathematics $A \cup B$ always includes the case where both happen. This is called the **non-exclusivity of or**, and it is the reason the addition rule below needs a correction term. When a question really does mean one or the other but not both, it will say so.

A **Venn diagram** makes the structure visible. The rectangle is the sample space U, each circle is an event, and every point of the diagram lies in exactly one of four regions. Combining an event with a complement names each region precisely.

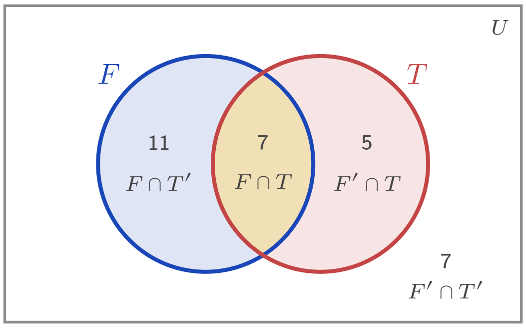

Figure 12.1 The four regions of a two-event Venn diagram, with the class of 30 filled in. Football only is

$F \cap T'$, both is $F \cap T$, tennis only is $F' \cap T$, and neither is $F' \cap T'$, the region outside both circles. The four counts add to 30.

Fill a Venn diagram from the middle outwards. Put the overlap in first, then subtract it from each circle total to get the "only" regions, then use the grand total to find the outside. Working the other way round double counts the overlap, which is the same mistake the addition rule is built to correct.

The addition rule

If you add $P(A)$ and $P(B)$ to find $P(A \cup B)$, you count the overlap twice: once in A, once in B. Subtracting it once corrects the double count.

KEY · IB

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

When two events cannot happen together, they are **mutually exclusive**: their intersection is empty, $P(A \cap B) = 0$, and the rule simplifies.

KEY · IB For mutually exclusive events:

$$P(A \cup B) = P(A) + P(B)$$

EXAMPLE 12.5

In a class of 30, 18 students play football, 12 play tennis, and 7 play both. A student is chosen at random. Find the probability that the student plays football or tennis, and the probability that they play neither.

Apply the addition rule directly:

$$P(F \cup T) = \frac{18}{30} + \frac{12}{30} - \frac{7}{30} = \frac{23}{30}.$$

"Neither" is the complement of "football or tennis":

$$P(\text{neither}) = 1 - \frac{23}{30} = \frac{7}{30}.$$

The Venn diagram above shows the same thing: 11 play only football, 7 play both, 5 play only tennis, and 7 play neither, totalling 30.

EXAMPLE 12.6

A single card is drawn from a standard deck of 52. Find the probability that it is a king or a heart.

There are 4 kings and 13 hearts, but the king of hearts is both, so it is counted in each:

$$P(\text{king} \cup \text{heart}) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}.$$

Forgetting to subtract the king of hearts gives $\frac{17}{52}$, which is the most common error in this topic.

Three events need three circles, and the same instruction applies: start in the middle. The centre region belongs to all three events, and every other region is found by subtracting what has already been placed.

EXAMPLE 12.7

In a year group of 40 students, 20 study French, 18 study German and 15 study Spanish. Of these, 8 study French and German, 6 study German and Spanish, 7 study French and Spanish, and 3 study all three. A student is chosen at random. Find the probability that the student studies (a) exactly one of the three languages, (b) none of them.

Start with the 3 in the centre. The 8 who study French and German include those 3, so $8-3 = 5$ study those two and no other. In the same way $6 - 3 = 3$ study German and Spanish only, and $7 - 3 = 4$ study French and Spanish only.

Now the "only" regions. French has 20 altogether, of which $5+3+4 = 12$ are already placed, leaving 8 for French only. German gives $18 - (5+3+3) = 7$, and Spanish gives $15 - (4+3+3) = 5$.

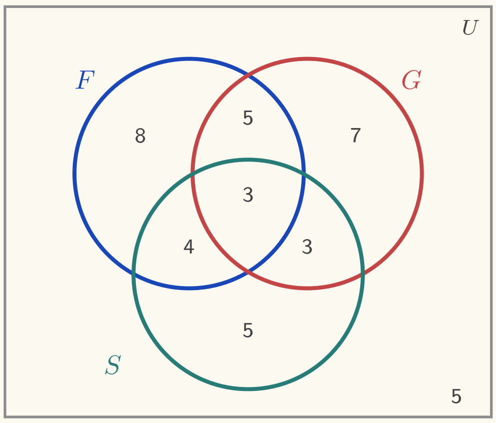

Figure 12.2 The three-circle diagram, filled from the centre outwards. The seven inner counts total 35, so 5 of the 40 students lie outside all three circles.

(a) Exactly one language means the three outer regions: $8 + 7 + 5 = 20$, so

$$P(\text{exactly one}) = \frac{20}{40} = \frac{1}{2}.$$

(b) The seven regions inside the circles total 35, so 5 students study none:

$$P(\text{none}) = \frac{5}{40} = \frac{1}{8}.$$

No formula was used. Once the diagram is filled, every probability is a count divided by 40.

Sample space diagrams and two-way tables

When an experiment has two parts, drawing the whole sample space is often faster than any formula. A **sample space diagram** plots one part on each axis. For two dice, the 36 equally likely outcomes form a grid, and any event is just a set of grid points to count.

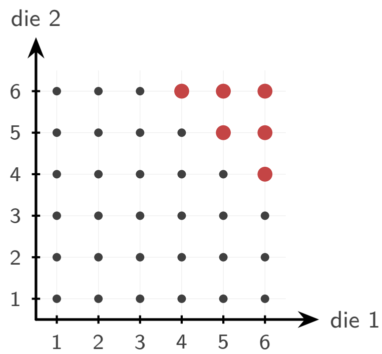

Figure 12.3 The 36 equally likely outcomes of rolling two dice. The six larger scarlet points are the outcomes whose total is at least 10.

The six larger scarlet points are the outcomes with a total of at least 10. There are six of them, so

$$P(\text{total} \geq 10) = \frac{6}{36} = \frac{1}{6}.$$

No rule was needed, only a picture and a count.

A **two-way table** (a table of outcomes) does the same for grouped data: it splits a population by two characteristics at once, and every probability is a cell, row, or column total divided by the grand total.

EXAMPLE 12.8

The table shows the 100 students of a year group by age group and club membership.

	Club	No club	Total
Junior	14	26	40
Senior	21	39	60
Total	35	65	100

A student is chosen at random. Find the probability that the student (a) is a junior in a club, (b) is in a club or is a junior.

(a) Read the single cell where the two conditions meet:

$$P(J \cap C) = \frac{14}{100} = 0.14.$$

(b) Apply the addition rule with the row and column totals:

$$P(J \cup C) = \frac{40}{100} + \frac{35}{100} - \frac{14}{100} = \frac{61}{100} = 0.61.$$

> The table already contains every count used in both calculations. We will return to this table in the next section, where it answers conditional questions just as directly.

YOUR TURN 12.2 Combining Events

5.

(a) Given $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$, find $P(A \cup B)$.
(b) Events A and B are mutually exclusive, with $P(A) = 0.6$ and $P(B) = 0.3$. Find $P(A \cup B)$.
(c) Given $P(A \cup B) = 0.8$, $P(A) = 0.5$ and $P(B) = 0.4$, find $P(A \cap B)$.
(d) Given $P(A) = 0.3$, $P(B) = 0.4$ and $P(A \cup B) = 0.7$, determine whether A and B are mutually exclusive.

6.

(a) A card is drawn from a deck of 52. Find the probability that it is a face card or a spade.
(b) In a group of 40 people, 25 like coffee, 20 like tea and 10 like both. Find the probability that a person chosen at random likes coffee or tea.
(c) Two events satisfy $n(A \text{ only}) = 8$, $n(B \text{ only}) = 5$, $n(A \cap B) = 4$ and $n(\text{neither}) = 3$. Find $P(A)$.
(d) Given $n(U) = 50$, $n(A) = 22$, $n(B) = 18$ and $n(A \cap B) = 7$, find $n(A \cap B')$ and $n(A' \cap B')$.
(e) In a group of 30 students, 15 play the piano, 12 the guitar and 9 the drums. Of these, 6 play piano and guitar, 4 play guitar and drums, 5 play piano and drums, and 2 play all three. Find the probability that a student chosen at random plays exactly one instrument, and the probability that they play none.

7. Two fair dice are rolled.

(a) Find the probability that the total is 7.
(b) Find the probability that the two numbers differ by exactly 1.
(c) Find the probability that the two dice show the same number.

8. The table shows 80 people classified by sex and by writing hand.

	Left	Right	Total
Male	7	33	40
Female	5	35	40
Total	12	68	80

One person is chosen at random.

(a) Find the probability that the person is left-handed.
(b) Find the probability that the person is female and right-handed.
(c) Find the probability that the person is female or left-handed.

12.3 Conditional Probability and Independence

Information changes probability. The chance that a random person has a particular illness is small, but the chance *given* that they tested positive is different. **Conditional probability** is the probability of one event once you know another has happened.

The essential idea is that knowing B has occurred **reduces the sample space** to B alone. Within that smaller set, you ask what fraction also lies in A.

KEY · IB

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) \neq 0$$

Read $P(A \mid B)$ as "the probability of A given B." The denominator $P(B)$ is the new, restricted sample space; the numerator is the part of it where A also holds.

EXAMPLE 12.9

Using the class of 30 from before, a student is chosen at random. Given that they play football, find the probability that they also play tennis.
Restrict to the 18 football players. Of those, 7 also play tennis:

$$P(T \mid F) = \frac{P(T \cap F)}{P(F)} = \frac{7/30}{18/30} = \frac{7}{18}.$$

Note this differs from $P(F \mid T) = \frac{7}{12}$. Conditioning is not symmetric: the order matters.

The multiplication rule

Rearranging the conditional formula gives a rule for the probability that two events both happen.

KEY

$$P(A \cap B) = P(A)\,P(B \mid A)$$

In words: the chance of both is the chance of the first times the chance of the second given the first. This is exactly how you move along the branches of a tree diagram, as the next section shows.

Independence

Two events are **independent** when knowing one tells you nothing about the other, so $P(A \mid B) = P(A)$. Substituting this into the multiplication rule gives a cleaner, symmetric test.

KEY · IB A and B are **independent** if and only if

$$P(A \cap B) = P(A)\,P(B),$$

or equivalently

$$P(A \mid B) = P(A) = P(A \mid B').$$

To test independence, compute both sides of either form and compare. The conditional form states directly what independence means: whether B happened or did not happen, the probability of A is unchanged.

EXAMPLE 12.10

A fair coin and a fair die are tossed together. Let H be "heads" and S be "rolling a six." Are H and S independent?

Here $P(H) = \frac{1}{2}$ and $P(S) = \frac{1}{6}$. The outcome "heads and a six" is one of the 12 equally likely combined outcomes:

$$P(H \cap S) = \frac{1}{12} = \frac{1}{2} \times \frac{1}{6} = P(H)\,P(S).$$

The two sides match, so the events are independent, which fits intuition: a coin cannot influence a die.

The two-way table of §12.2 passes the same test. There, $P(C \mid J) = \frac{14}{40} = 0.35$ and $P(C) = \frac{35}{100} = 0.35$: knowing a student is a junior does not change the chance they are in a club, so age group and club membership are independent.

CAUTION *Mutually exclusive and independent are opposites, not synonyms. If two events with non-zero probability are mutually exclusive, then one happening prevents the other completely, which is the strongest possible dependence. Never assume an "or" situation is also an "independent" one.*

YOUR TURN 12.3 Conditional Probability and Independence

9.

(a) Given $P(A \cap B) = 0.2$ and $P(B) = 0.5$, find $P(A \mid B)$.
(b) Given $P(A) = 0.6$ and $P(B \mid A) = 0.3$, find $P(A \cap B)$.
(c) Given $P(A \mid B) = 0.7$ and $P(B) = 0.4$, find $P(A \cap B)$.
(d) Two fair dice are rolled. Given that the first die shows an even number, find the probability that the total is 8.

10.

(a) Events A and B are independent, with $P(A) = 0.5$ and $P(B) = 0.4$. Find $P(A \cap B)$.
(b) Given $P(A) = 0.3$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$, determine whether A and B are independent.
(c) Two cards are drawn from a deck of 52 without replacement. Find the probability that both are kings.

11. In a class of 25 students, 15 study Spanish, 10 study art and 6 study both.

(a) Given that a student studies Spanish, find the probability that they also study art.
(b) Given that a student studies art, find the probability that they also study Spanish.
(c) Determine whether studying Spanish and studying art are independent.
(d) A card is drawn from a deck of 52. Given that it is a face card, find the probability that it is a king.

12.4 Tree Diagrams

When events happen in sequence, a **tree diagram** organises every path. Each branch carries the probability of that step, and two rules then do the whole calculation: **multiply along the branches** of a single path (the multiplication rule), and **add across paths** that satisfy the condition (the paths are mutually exclusive).

When each draw is made **with replacement**, the object goes back before the next draw, every stage repeats the same probabilities, and the stages are independent. Care is needed with sequences **without replacement**, where the first outcome changes the probabilities on the second branches.

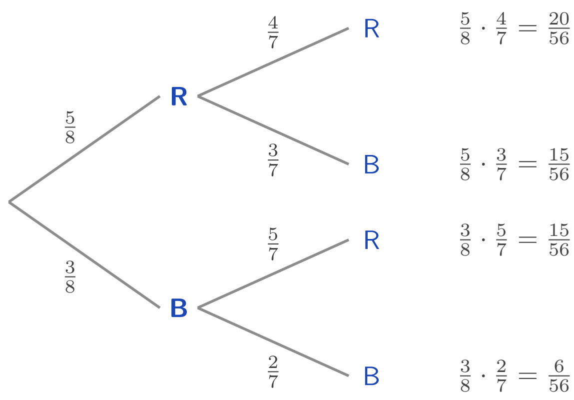

Figure 12.4 A tree diagram for drawing two marbles from 5 red and 3 blue without replacement. The probabilities on the second stage differ from the first, because one marble has already been removed. Multiply along a path; add across paths.

EXAMPLE 12.11

A bag holds 5 red and 3 blue marbles. Two are drawn without replacement. Find the probability that (a) both are red, (b) one of each colour is drawn, (c) at least one is blue.

After drawing one red, only 4 reds remain out of 7 marbles, which is why the second branch changes.

(a) Multiply along the top path:

$$P(\text{both red}) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}.$$

(b) "One of each" covers two paths, red-then-blue and blue-then-red; add them:

$$P(\text{one of each}) = \frac{5}{8} \cdot \frac{3}{7} + \frac{3}{8} \cdot \frac{5}{7} = \frac{15+15}{56} = \frac{30}{56} = \frac{15}{28}.$$

(c) Use the complement of "at least one blue," which is "both red":

$$P(\text{at least one blue}) = 1 - \frac{5}{14} = \frac{9}{14}.$$

All three rules appear here: multiply along a path, add across alternatives, and take the complement for "at least one."

Putting the marble back changes every second-stage branch. The bag returns to 5 red and 3 blue, so both stages read $\frac{5}{8}$ and $\frac{3}{8}$, and the two draws are independent.

EXAMPLE 12.12

The same bag holds 5 red and 3 blue marbles, but now the first marble is replaced before the second is drawn. Find the probability that both are red, and compare with the answer without replacement.

Every branch keeps its original probability, so

$$P(\text{both red}) = \frac{5}{8} \times \frac{5}{8} = \frac{25}{64} \approx 0.391.$$

Without replacement the answer was $\frac{5}{14} \approx 0.357$, which is smaller. Removing a red marble leaves fewer reds for the second draw, so the second red becomes less likely. Replacing it removes that effect.

Replacement also makes the most useful pattern in this chapter available. When the same trial is repeated n times independently, "at least one success" has a complement that is a single product, because "no successes at all" means failure every time.

KEY For n independent trials in which the event has probability p each time,

$$P(\text{at least one}) = 1 - (1-p)^n.$$

EXAMPLE 12.13

A machine part fails on any given day with probability 0.02, independently of other days. Find the probability that it fails at least once during a 30-day month.

Counting the ways to fail once, twice, three times and so on would take all day. The complement is a single product: the part survives all 30 days, each day with probability 0.98.

$$P(\text{at least one failure}) = 1 - (0.98)^{30} = 1 - 0.545 = 0.455.$$

A part that is safe on 98 days out of 100 still has close to an even chance of failing within a month. A small probability repeated often stops being small, which is why maintenance is scheduled by the month and not by the day.

YOUR TURN 12.4 Tree Diagrams

12.

(a) A bag holds 3 white and 2 black balls. Two are drawn with replacement. Find the probability that both are white.

(b) Repeat part (a) for two balls drawn without replacement.

(c) A box holds 6 working and 4 defective bulbs. Two are drawn without replacement. Find the probability that exactly one is defective.

13.

(a) A machine works on any given day with probability 0.9, independently of other days. Find the probability that it works on two consecutive days.

(b) The probability of rain is 0.3. If it rains, the probability of being late is 0.6; if it does not, the probability is 0.1. Find the probability of being late.

(c) A fair die is rolled three times. Find the probability of at least one six.

(d) Using the machine in part (a), find the probability that it fails on at least one of five consecutive days.

14.

(a) A bag holds 4 red and 6 blue counters. Three are drawn without replacement. Find the probability that all three are blue.
(b) For the same three draws, find the probability that exactly two are red.
(c) A component works with probability 0.95, independently of other components. Find the probability that all four components in a circuit work.
(d) Find the probability that at least one of the four components fails.

12.5 Bayes' Theorem HL

We can now return to the letter. The difficulty is that the test gives you $P(\text{positive} \mid \text{disease})$, but what you actually want is the reverse, $P(\text{disease} \mid \text{positive})$. Conditional probability is not symmetric, so these are different numbers. **Bayes' theorem** is the tool that reverses the condition.

It is built from two facts you already have. The conditional formula gives $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$, and the denominator can be split according to whether A happened or not: $P(B) = P(A)P(B \mid A) + P(A')P(B \mid A')$. Putting these together:

KEY · IB HL

$$P(A \mid B) = \frac{P(A)\,P(B \mid A)}{P(A)\,P(B \mid A) + P(A')\,P(B \mid A')}$$

The numerator is the single path through A; the denominator is every path that leads to B. You are asking what fraction of all the ways to observe B passed through A.

That description is the key to using it. Written out in full the formula is long, but a tree diagram carries the same information in a form that is quicker to set up. Draw the first stage, A and A'. Draw the second stage, B given each of them. Multiply along every path. Then divide the path that passes through A by the total of all the paths that end in B.

TIP *Do not memorise the Bayes formula as a string of symbols. Draw the tree, multiply along the paths, and divide the path you want by the total of the paths that match the evidence. It gives the same answer, it is far harder to get wrong under exam pressure, and it earns the same marks.*

EXAMPLE 12.14

A disease affects 1% of a population. A test is positive for 99% of those who have it, and (falsely) positive for 5% of those who do not. A person tests positive. Find the probability they actually have the disease.

Let D be "has the disease" and $+$ be "tests positive." We know $P(D) = 0.01$, $P(+ \mid D) = 0.99$, and $P(+ \mid D') = 0.05$. Apply Bayes:

$$P(D \mid +) = \frac{(0.01)(0.99)}{(0.01)(0.99) + (0.99)(0.05)} = \frac{0.0099}{0.0099 + 0.0495} = \frac{0.0099}{0.0594} \approx 0.167.$$

The same calculation reads straight off a tree. The upper path gives $0.01 \times 0.99 = 0.0099$, and the lower path gives $0.99 \times 0.05 = 0.0495$. Divide the upper path by the total of both.

Only about 17%. The reason is the base rate. The disease is so rare that the few true positives are greatly outnumbered by the false positives, which come from the very large healthy majority. The test is good; the disease is simply very rare.

Three alternatives

When the first stage has three possibilities instead of two, the denominator simply sums over all three paths. The structure is identical.

EXAMPLE 12.15

Three machines produce all the parts in a factory: machine A makes 50%, B makes 30%, C makes 20%. Their defect rates are 1%, 2%, and 3% respectively. A part is found to be defective. Find the probability it came from machine C.

Weight each machine's defect rate by its share of production, then take C's share of the total:

$$P(C \mid \text{def}) = \frac{(0.20)(0.03)}{(0.50)(0.01) + (0.30)(0.02) + (0.20)(0.03)} = \frac{0.006}{0.005 + 0.006 + 0.006} = \frac{0.006}{0.017} \approx 0.353.$$

Machine C makes only a fifth of the parts but, because it is the least reliable, accounts for over a third of the defects.

YOUR TURN 12.5 Bayes' Theorem HL

15.

(a) Using the situation in question 13 part (b), a person is late. Find the probability that it rained.

(b) A disease affects 2% of a population. A test is positive for 95% of those who have it and for 10% of those who do not. Given a positive result, find the probability of having the disease.

(c) Factory A makes 60% of all items, of which 2% are defective; factory B makes the other 40%, of which 5% are defective. Given that an item is defective, find the probability it came from factory B.

16.

(a) In a mailbox, 30% of messages are spam. The word "free" appears in 90% of spam messages and in 5% of the others. A message contains the word "free". Find the probability that it is spam.
(b) A box contains two coins: one fair, and one with heads on both sides. A coin is chosen at random and tossed once, showing heads. Find the probability that it is the two-headed coin.
(c) Of the students in a year group, 70% revise for a test. Of those who revise, 80% pass; of those who do not, 30% pass. A student passes. Find the probability that the student revised.

17. Three suppliers provide the components used by a factory. Supplier A provides 50%, supplier B 30% and supplier C 20%. Of their components, 1%, 3% and 4% respectively are defective.

(a) Find the probability that a component chosen at random is defective.
(b) Given that a component is defective, find the probability that it came from supplier B.
(c) Given that a component is defective, find the probability that it came from supplier A.
(d) Supplier A provides the most components but accounts for the fewest defective ones. Explain why.

Chapter 12 · Probability

Reflections

IM Probability began at a gambling table. In 1654 the French writer Antoine Gombaud, the Chevalier de Méré, asked Blaise Pascal why a betting strategy that seemed sound was losing him money. Pascal's correspondence with Pierre de Fermat over that single question laid the foundations of probability theory. A subject now central to physics, finance, and medicine grew out of a dispute about dice.

TOK Bayes' theorem describes how a rational mind should update a belief when new evidence arrives: start from a *prior*, the probability believed before the evidence, weigh the evidence, and arrive at a *posterior*, the probability believed afterwards. Some philosophers argue this is not just a formula but a model of knowledge itself. If that is right, then the disease-test paradox is not a quirk of medicine. It is a warning that evidence means little until you account for how rare the thing you are testing for actually is.

TOK We say a fair die has probability $\frac{1}{6}$ because of its symmetry, and we say a drawing pin lands point-up with probability 0.39 because that is what 200 drops showed. The first number was never measured; the second was never proved.

Are these two the same kind of claim? And when a measured value disagrees with a calculated one, which of the two should be revised?

TOK Probability was built to answer questions about gambling, and casinos use it now to guarantee themselves a profit. The mathematics itself is neutral, but the uses of it are not.

Should a mathematician who works out the odds be held responsible for how those odds are used?

RW The base-rate paradox has real costs. Courtrooms have convicted people on "one-in-a-million" DNA matches without asking how many people in the population would also match by chance. Mass screening programmes for rare cancers can produce more false alarms than true detections. Understanding $P(D \mid +)$ versus $P(+ \mid D)$ is not only an examination topic. It changes verdicts and medical decisions.

End-of-Chapter Problems — Chapter 12: Probability

1. In a class of 50 students, 28 study Mathematics (M), 20 study Art (A), and 10 study neither.

(a) Find the number of students who study both Mathematics and Art. [3]
(b) Find the probability that a randomly chosen student studies only Art. [2]
(c) Find $P(A \mid M)$. [2]

2. The two-way table shows whether 100 people exercise regularly.

	Exercises	Does not	Total
Male	30	20	50
Female	24	26	50
Total	54	46	100

(a) Find the probability that a person is male and exercises. [2]
(b) Find the probability that a person exercises, given that they are female. [2]
(c) Determine whether the events "male" and "exercises" are independent. [3]

3. A bag contains 7 red and 3 green sweets. Two are drawn without replacement.

(a) Draw a tree diagram for the two draws. [3]
(b) Find the probability that both sweets are red. [2]
(c) Find the probability that both sweets are the same colour. [3]
(d) Find the probability that at least one sweet is green. [2]

4. A condition affects 5% of a population. A screening test is positive for 90% of those who have it, and for 15% of those who do not.

(a) Find the probability that a randomly chosen person tests positive. [3]
(b) Given a positive result, find the probability the person has the condition. [3]
(c) Comment on what your answer says about screening for rare conditions. [2]

5. Events A and B satisfy $P(A) = 0.5$ and $P(B) = 0.3$.

(a) Find $P(A \cup B)$ if A and B are independent. [3]
(b) Find $P(A \cup B)$ if A and B are mutually exclusive. [2]
(c) Explain why A and B cannot be both independent and mutually exclusive here. [2]

6. Two fair dice are rolled and the scores noted.

(a) Find the probability that the sum is 7. [2]

(b) Find the probability that at least one die shows a 6. [2]
(c) Given that at least one die shows a 4, find the probability that the sum is 7. [3]

7. Three suppliers provide components: X supplies 25%, Y supplies 45%, and Z supplies 30%. Their defect rates are 4%, 2%, and 5% respectively.

(a) Find the probability that a randomly chosen component is defective. [3]
(b) Given that a component is defective, find the probability it came from Z. [3]

8. Events A and B are independent. Given $P(A) = 0.4$ and $P(A \cup B) = 0.7$, find $P(B)$. [4]

9. A basketball player makes each free throw independently with probability 0.8. She takes 4 free throws.

(a) Find the probability she makes all four. [2]
(b) Find the probability she misses at least one. [2]

10. At a school, 60% of students study French and 25% study both French and Spanish. Find the probability that a student studies Spanish, given that they study French. [3]

11. In a class of 40 students, the numbers studying two activities A and B are: only A is $2x$, both is x, only B is $x + 4$, and neither is 8.

(a) Find the value of x. [3]
(b) Find $P(A \cap B)$. [1]
(c) Find $P(A \cup B)$. [2]

12. A biased four-sided spinner numbered 1 to 4 was spun 250 times. The results were:

Number	1	2	3	4
Frequency	95	60	55	40

(a) Estimate the probability that the spinner lands on 1. [2]
(b) Estimate the probability that the spinner lands on an odd number. [2]
(c) The spinner is spun a further 60 times. Find the expected number of times it lands on 4. [2]

13. Events A and B satisfy $P(A) = 0.45$, $P(B) = 0.4$, and $P((A \cup B)') = 0.25$.

(a) Find $P(A \cap B)$. [2]
(b) Find the probability that A occurs but B does not. [1]
(c) Find $P(A \mid B)$. [2]

(d) Determine whether A and B are independent. [2]

14. A fair four-sided die (numbered 1–4) and a fair six-sided die (numbered 1–6) are rolled together, and the product of the two scores is recorded.

(a) Represent the outcomes in a sample space diagram. [2]
(b) Find the probability that the product is 12. [2]
(c) Find the probability that the product is odd. [2]
(d) Given that the product is even, find the probability that the four-sided die shows a 4. [3]

15. A football team's key striker is fit for a match with probability 0.85. The team wins with probability 0.7 if she is fit, and with probability 0.4 if she is not.

(a) Draw a tree diagram to represent this information. [2]
(b) Find the probability that the team wins the match. [2]
(c) Given that the team won, find the probability that the striker was fit. [3]

16. A box contains 4 red and 2 green counters. Three counters are drawn at random without replacement.

(a) Find the probability that all three counters are red. [2]
(b) Find the probability that exactly one counter is green. [3]
(c) Find the probability that at least one counter is green. [2]

17. Events A and B satisfy $P(A) = 0.6$, $P(A \mid B) = 0.6$, and $P(A \cup B) = 0.84$.

(a) Explain why A and B are independent. [1]
(b) Find $P(B)$. [3]
(c) Write down $P(B \mid A')$, justifying your answer. [2]

18. The 80 students of a year group are classified by gender and handedness: 10 of the 40 boys and 6 of the 40 girls are left-handed.

(a) Display this information in a two-way table, including totals. [2]
(b) Find the probability that a randomly chosen student is a right-handed girl. [1]
(c) Given that a student is left-handed, find the probability that the student is a boy. [2]
(d) Determine, using conditional probability, whether the events "boy" and "left-handed" are independent. [3]

19. An airline routes 50% of its flights through airport A, 30% through airport B, and 20% through airport C. The probabilities that a flight is delayed are 0.06, 0.03, and 0.10 respectively.

(a) Find the probability that a randomly chosen flight is delayed. [3]
(b) Given that a flight was delayed, find the probability that it was routed through airport A. [3]

20. Events A, B, and C are mutually exclusive and exhaustive, with $P(A) = 2k$, $P(B) = 3k$, and $P(C) = k + 0.1$.

(a) Find the value of k. [2]
(b) Find $P(A \cup B)$. [2]
(c) Find $P(A \mid B')$. [3]

21. GDC An archer hits the target with probability 0.9 on each shot, independently of all other shots.

(a) Find the probability that her first miss is on her third shot. [2]
(b) Find the probability that she misses at least once in five shots. [2]
(c) Find the least number of shots for which the probability of at least one miss exceeds 0.95. [3]

22. A family has two children. Assume each child is equally likely to be a boy or a girl, independently.

(a) List the sample space. [1]
(b) Given that at least one of the children is a boy, find the probability that both are boys. [3]
(c) Given instead that the *elder* child is a boy, find the probability that both are boys. [2]
(d) Explain why the answers to (b) and (c) differ. [1]

23. In a survey of 100 people, 50 like tea, 60 like coffee, and 35 like juice. Also, 30 like tea and coffee, 20 like tea and juice, 25 like coffee and juice, and 10 like all three.

(a) Find the number who like at least one of the three drinks. [3]
(b) Find the number who like none. [1]
(c) Find the number who like exactly one drink. [3]

24. A disease affects 1% of a population. A test is positive for 99% of those with the disease and for 5% of those without. A person tests positive *twice*, the two tests being independent given disease status.

(a) Find the probability the person has the disease after one positive test. [3]
(b) Find the probability the person has the disease after two positive tests. [4]
(c) Comment on the effect of the second test. [1]

25. Events A and B are independent. Prove that A and B' are also independent, that is, show $P(A \cap B') = P(A)\,P(B')$. [5]

26. From a group of 5 men and 4 women, three people are chosen at random. Find the probability that exactly two are women. [5]

27. Of the 200 students at a school, 130 cycle to school (C), 90 travel by train (T), and 40 do both. The remaining students walk.

(a) Draw a Venn diagram showing the number of students in each of the four regions. [3]
(b) Find the probability that a student chosen at random walks to school. [2]
(c) Find $P(C \mid T)$. [3]
(d) Determine whether C and T are independent, justifying your answer. [3]
(e) Two students are chosen at random, without replacement. Find the probability that both of them walk. [3]
(f) State whether your answer to part (e) would be larger or smaller if the two students were chosen with replacement, and explain why. [2]

28. A machine produces components, of which 6% are defective. Each component is inspected. The inspection rejects a defective component with probability 0.95, and wrongly rejects a component that is not defective with probability 0.02.

(a) Draw a tree diagram to represent this information. [3]
(b) Find the probability that a randomly chosen component is rejected. [3]
(c) Find the probability that a rejected component was in fact not defective. [3]
(d) Find the probability that a component which passes inspection is defective. [3]
(e) In a batch of 5000 components, find the expected number that are defective but pass inspection. [2]
(f) Using your answers to parts (d) and (e), comment on how well the inspection protects a customer. [2]

29. A box contains 10 batteries, of which 3 are flat. Batteries are taken from the box one at a time, without replacement, and tested until a working one is found. Let N be the number of batteries tested.

(a) Find $P(N = 1)$. [2]
(b) Find $P(N = 2)$. [3]
(c) Find $P(N \leq 3)$. [4]
(d) Explain why N can never be greater than 4. [2]
(e) Find $P(N = 3)$ and $P(N = 4)$, and verify that the four probabilities sum to 1. [4]

Challenge Questions

30. The birthday problem. In a group of n people, assume every birthday is equally likely among 365 days and ignore leap years.

(a) For $n = 2$, show that the probability the two birthdays differ is $\dfrac{364}{365}$. [2]

(b) Explain why, for $n = 3$, the probability that all three differ is $\dfrac{364}{365} \times \dfrac{363}{365}$. [3]

(c) Write down an expression for the probability that all n birthdays differ. [3]

(d) Hence find, to three decimal places, the probability that at least two people share a birthday when $n = 10$, $n = 23$ and $n = 50$. [4]

(e) Find the smallest n for which the probability of a shared birthday exceeds $\frac{1}{2}$. [3]

(f) Most people guess that n must be close to 180. Explain why the true answer is so much smaller, by considering how many *pairs* of people a group of n contains. [3]

31. Three doors. A prize is behind one of three closed doors. You choose a door. The host knows where the prize is, and opens one of the other two doors, always an empty one. You may keep your door or switch to the last closed door.

(a) Write down the probability that your first choice is correct. [1]

(b) Find the probability of winning if you always switch. Treat the case where your first choice is correct separately from the case where it is not. [5]

(c) State the probability of winning if you always stay, and say which strategy is better. [2]

(d) The host always opens an empty door, so his action seems to tell you nothing. Explain why it does. [4]

(e) The game is now played with n doors and one prize, and the host opens every door except your choice and one other. Show that switching wins with probability $\dfrac{n-1}{n}$. [4]

32. The Chevalier's puzzle. In 1654 a French gambler won money on one of these two bets and lost money on the other. He could not see why.

(a) Bet 1: roll one die four times, and win if at least one six appears. Show that the probability of winning is $\dfrac{671}{1296}$. [4]

(b) Bet 2: roll a pair of dice 24 times, and win if at least one double six appears. Find this probability, to four decimal places. [4]

(c) State which bet the gambler should take, and explain why the two probabilities are so close. [3]

(d) He argued that 24 rolls of two dice must be as good as 4 rolls of one, because 24 is to 36 as 4 is to 6. Explain the error. [3]

(e) Find the smallest number of rolls of a pair of dice for which Bet 2 is favourable. [4]